Cognizant ACE DSA Question List - Pattern Based Preparation

Arrays

No.	Question	Pattern	Done
1	Two Sum	Arrays	■
2	Best Time to Buy and Sell Stock	Arrays	■
3	Maximum Subarray (Kadane's Algorithm)	Arrays	■
4	Move Zeroes	Arrays	■
5	Remove Duplicates from Sorted Array	Arrays	■
6	Rotate Array	Arrays	■
7	Majority Element	Arrays	■
8	Missing Number	Arrays	■
9	Find Duplicate Number	Arrays	■
10	Merge Sorted Array	Arrays	■
11	Intersection of Two Arrays	Arrays	■
12	Sort Colors	Arrays	■
13	Maximum Product Subarray	Arrays	■
14	Longest Consecutive Sequence	Arrays	■
15	Merge Intervals	Arrays	■

Strings

No.	Question	Pattern	Done
16	Valid Palindrome	Strings	■
17	Valid Anagram	Strings	■
18	Reverse String	Strings	■
19	Reverse Words in a String	Strings	■
20	Longest Common Prefix	Strings	■
21	First Non-Repeating Character	Strings	■
22	Longest Substring Without Repeating Characters	Strings	■
23	String Compression	Strings	■
24	Check String Rotation	Strings	■
25	Implement strstr / Pattern Matching	Strings	■

Hashing

No.	Question	Pattern	Done
26	Frequency Count of Elements	Hashing	■
27	Two Sum using HashMap	Hashing	■
28	Subarray Sum Equals K	Hashing	■
29	Longest Consecutive Sequence	Hashing	■
30	Group Anagrams	Hashing	■

Sliding Window

No.	Question	Pattern	Done
31	Maximum Sum Subarray of Size K	Sliding Window	■
32	Longest Substring with K Distinct Characters	Sliding Window	■
33	Minimum Window Substring	Sliding Window	■
34	Longest Repeating Character Replacement	Sliding Window	■

Binary Search

No.	Question	Pattern	Done
35	Binary Search	Binary Search	■
36	First and Last Position of Element	Binary Search	■
37	Search Insert Position	Binary Search	■
38	Search in Rotated Sorted Array	Binary Search	■
39	Find Peak Element	Binary Search	■
40	Square Root using Binary Search	Binary Search	■

Linked List

No.	Question	Pattern	Done
41	Reverse Linked List	Linked List	■
42	Find Middle of Linked List	Linked List	■
43	Detect Cycle in Linked List	Linked List	■
44	Merge Two Sorted Linked Lists	Linked List	■
45	Remove Nth Node From End	Linked List	■
46	Palindrome Linked List	Linked List	■

Stack / Queue

No.	Question	Pattern	Done
47	Valid Parentheses	Stack / Queue	■

No.	Question	Pattern	Done
48	Implement Stack using Queue	Stack / Queue	■
49	Implement Queue using Stack	Stack / Queue	■
50	Next Greater Element	Stack / Queue	■
51	Min Stack	Stack / Queue	■
52	Largest Rectangle in Histogram	Stack / Queue	■

Trees

No.	Question	Pattern	Done
53	Binary Tree Traversals	Trees	■
54	Maximum Depth of Binary Tree	Trees	■
55	Level Order Traversal	Trees	■
56	Check Balanced Binary Tree	Trees	■
57	Diameter of Binary Tree	Trees	■
58	Lowest Common Ancestor	Trees	■
59	Validate Binary Search Tree	Trees	■
60	Search in Binary Search Tree	Trees	■

Recursion / Backtracking

No.	Question	Pattern	Done
61	Fibonacci using Recursion	Recursion / Backtracking	■
62	Generate Subsets	Recursion / Backtracking	■
63	Generate Permutations	Recursion / Backtracking	■
64	Combination Sum	Recursion / Backtracking	■

Dynamic Programming

No.	Question	Pattern	Done
65	Climbing Stairs	Dynamic Programming	■
66	House Robber	Dynamic Programming	■
67	Coin Change	Dynamic Programming	■
68	Longest Increasing Subsequence	Dynamic Programming	■
69	Longest Common Subsequence	Dynamic Programming	■